DATA ENGINEERING

ELT vs ETL Visual Guide

Where does the transform happen?
The pipeline choice that shapes modern data stacks.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Classic ETL Pipeline

Extract → Transform → Load. Transform happens BEFORE the warehouse.

ETL transforms in a separate engine (Informatica, SSIS). The warehouse only ever sees polished, schema-fixed data.

- Raw data is discarded or archived before loading
- Schema must be decided upfront (schema-on-write)
- Transform engine is a separate cluster to buy and maintain

STEP 2/6

ELT: Flip the Order

Extract → Load → Transform. Land raw data first, transform inside the warehouse.

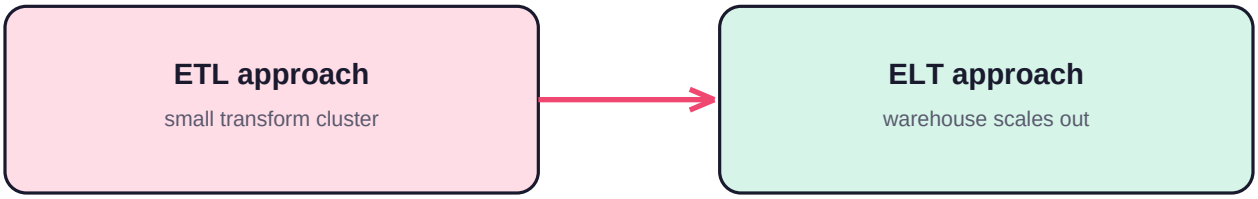
ELT lands raw data first (bronze), then transforms with warehouse compute into silver/gold layers.

- Raw data is always preserved — re-transform anytime
- Schema-on-read: decide the shape when you query it
- One system to maintain instead of two

STEP 3/6

Pushdown: The Warehouse Does the Work

Cloud warehouses have massive elastic compute — use it.



Transform engine sized for peak load, idle most of the time = wasted spend.

Transform runs where the data lives, on elastic compute that scales to zero.

Pushdown compute means no data movement for transforms — the engine comes to the data, not the other way around.

STEP 4/6

When Each Wins

No universal winner — match the pattern to the constraint.

ETL wins

- Heavy PII scrubbing before storage
- Regulated data that must never land raw
- Legacy on-prem warehouses (Teradata, Oracle)

ELT wins

- Exploratory analytics, ML features
- Re-processing history with new logic
- Cloud warehouses (Snowflake, BigQuery)

STEP 5/6

ELT in Practice: Medallion Layers

Raw → cleaned → business-ready, all inside the platform.

The medallion (bronze/silver/gold) architecture is ELT's signature: every stage is just SQL on warehouse tables.

- Bronze: exact copy of source, partitioned by ingestion time
- Silver: one row per entity, types fixed, PII tagged
- Gold: star schemas and aggregates the business queries

STEP 6/6

The Decision Framework

Ask these questions before picking a pattern.

- Must raw data NEVER be stored? → ETL (or ELT with in-flight masking)
- Do analysts need to re-shape history? → ELT keeps raw forever
- Is your warehouse elastic cloud? → ELT uses pushdown compute
- Small team, one stack? → ELT is simpler to operate

Modern default: ELT on a cloud warehouse, with ETL-style scrubbing only for data that legally cannot land raw.

STEP ✓/✓

Key Takeaways

ELT vs ETL — Remember

- ETL transforms before loading; ELT loads raw first and transforms inside the warehouse.
- ELT's superpower: raw data is preserved, so history can be re-transformed with new logic.
- Pushdown compute runs transforms on elastic warehouse capacity — no separate engine to size.
- ETL still wins when regulations forbid storing raw data (PII scrubbing before landing).
- Medallion layers (bronze/silver/gold) are the standard ELT shape for raw → clean → business-ready.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev